

Introduction to Software Project Management

Syllabus

Introduction, Why is Software Project Management Important ? What is a Project ? Software Projects versus Other Types of Project, Contract Management and Technical Project Management, Activities Covered by Software Project Management, Plans, Methods and Methodologies, Some Ways of Categorizing Software Projects, Project Charter, Stakeholders, Setting Objectives, The Business Case, Project Success and Failure, What is Management ? Management Control, Project Management Life Cycle, Traditional versus Modern Project Management Practices.

1.1 Introduction to Project Management

Software Project Management (SPM) is an effort taken to develop a unique software product or service. It involves :

- Definite timeframe
- Budget
- Definite specifications
- Working in organizational boundaries

Software Project Management is :

- Creating a favorable environment to get critical projects done.
- Ensuring that software is delivered on time and in accordance with the requirements of the developing organization as well as procuring organization – This includes defining, scoping, planning, staffing, organizing, directing, monitoring and controlling a project to develop an acceptable system within the allotted time and budget.
 - o It is an activity that documents, teaches, oversees the use and improves an organization's chosen methodology for system development.
 - o Consistently applying phases, activities, deliverables and quality standards to the projects.
 - o Applying tools/techniques, knowledge and skills to activities to meet the requirements of the project as well as to meet the stakeholder expectations.

1.1.1 Why is Software Project Management Important ?

Software project management is needed to manage business and environment risks that occur during project development process :

- Many software products are developed according to client's requirements so they are only useful to those specific clients.
- The software products which are developed for business and used by business clients become outdated due to the introduction of newer and advanced technology in business. As the technology advances frequently, the product developed in a particular technology may no cope-up with new requirements and changes and therefore, it becomes outdated after some decades.

- Goal of Project Management : Developing Project within budget, time and scope.

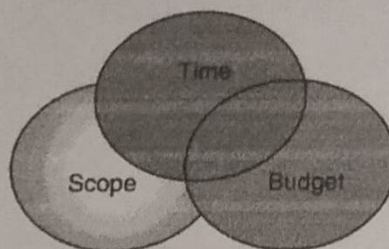


Fig. 1.1.1 : Goal of project management

1.2 What is a Project ?

Project :

- A project is a series of tasks that is carefully planned in order to achieve a particular outcome. Project can range from a simple to complex and can be managed by either individual or a team(s).
- Example: Planning a wedding party is a project because it is a specific type of party to be held on a specific date and time which needs to be accomplished by proper planning and certain inputs and the party gives a specific service.
- A project requires :
 - o Proper planning
 - o Specific objectives
 - o A timeframe
 - o Constrained resources

Software Project :

- Software project is a balance of three dimensions – time, features and resources with a goal of producing a successful software deliverable.
- To take up a new project, you need to take the risk and make an investment in order to get a positive payoff.
- Project Management supports three basic levels of projects :
 1. Projects
 2. Subprojects and
 3. Programs.

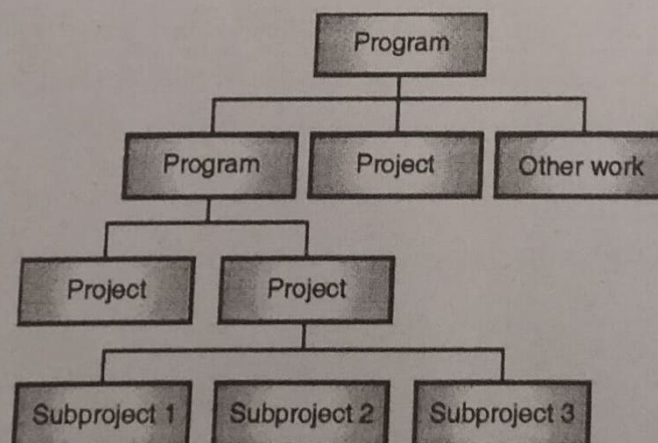


Fig. 1.2.1 : Project management levels



1. **A project** : It ranges from simple to large that require much time, effort, and money.
2. **A sub project** : It refers to a part of a project managed with a level of independence.
3. **A program** : It is a group of related projects and other activities managed in a coordinated way to achieve a common long-term objective.

Difference between a Program and a Project :

Table 1.2.1 : Difference between Project and Program

Sr. No	Project	Program
1.	Has a start and a finish	Consists of multiple projects with no start or finish.
2.	Has a set of defined deliverables	Scope can be very broad or very specific.
3.	Supports a tactical initiative	Supports a strategy or ongoing mission.
4.	Example : Updating all the computers in the hotel with one new version of software	Example : The ongoing IT support of all the computers in the hotel.

- Effective project management focuses on 4 P's i.e. the **People**, the most important element of a successful project, the **Product**, which has to be built, the **Process**, the set of tasks and activities to be executed, the **Project**, the combined work which makes the product a reality.

1.2.1 Software Projects v/s. Other Types of Projects

There are variety of projects apart from software project.

- **Construction** : These projects deal with civil or architectural work. Predictive methods are used along with agile techniques.
- **IT/Software** : These projects deal with software development, IT system etc. A SDLC model is used along with agile techniques.
- **Business** : These projects deal with the development of a business, management of a work team, cost management, etc. A commercial strategy is usually followed.
- **Service or Product production** : These projects deal with the development of an innovative product or service, or design of a new product, etc. such type of projects are often built in the R & D department.

Table 1.2.2 : Software Project v/s. Other Projects

Sr. No.	Software Project	Other types of Project
1.	Complexity : Testing the quality of software is difficult since so many paths and so many test cases to test.	Complexity : Testing the quality is comparatively easier. For example, bridges and buildings where everything is tested using known procedures.
2.	Invisibility : Progress in software development is not immediately visible	Invisibility : Progress in construction of a bridge or building can actually be seen.

Sr. No.	Software Project	Other types of Project
3.	Conformity : The ready software product should conform to the requirements of the clients.	Conformity : The ready product such as a bridge or building should confirm to the physical laws.
4.	Flexibility : When a software is migrated from one physical interface to another, it is expected that the software will change to accommodate the system.	Flexibility : Difficult to make any changes in a built product such as difficult to make any type of changes in an already built building.
5.	Highly market driven and shorter release cycles	There is no concept of release cycles
6.	Much higher rate of failure due to immaturity of the discipline.	Rate of failure is much less. There is no much variance in the inputs and procedures used to build a product.
7.	High rate of technological change	Comparatively low rate of technological change
8.	Software is easier to copy and hack	Other types of projects are difficult to copy

1.3 Contract Management and Technical Project Management

Sr. No.	Technical Project Management	Contract Management
1.	Technical Project management is about planning and executing the project based on scope, time and budget. It is about managing all aspects of the project.	Contract management is a subcategory of Project management which takes care of the legal aspects related to the project such as contracts with vendors, leases and licensing agreements.
2.	Focuses on meeting the project goals such as product launches, building projects, and software development.	It takes care of legal documents whenever the project work is contracted out. It ensures that the terms and commitments that are agreed upon in the contract are adhered to during the project development process. Offer by the company + customer reacts with changes to the offer + new offer by customer + acceptance by company and the customer = final contract agreement.
3.	Project manager meets the clients and works with them to build a plan	Contract managers help clients to understand the agreement they sign. They even work with the legal and financial departments of the organization to ensure the contract meets the needs of the organization.

Sr. No.	Technical Project Management	Contract Management
4.	Activities of a Technical Project Manager : - To develop a document that outlines project details - To report any delays in progress to the client - To work with vendors and third-party resources to outsource if required. - To meet all the standards and legal regulations and codes.	Activities of a Contract Manager : - To research on competitive companies with similar projects - To manage the clerical personnel responsible for filling and editing paperwork (agreements). - To negotiate with stakeholders to ensure a win-win contract. - To explain the risks (if any) to clients based on points in the agreement.

1.4 Software Project Management Activities

Activities involved in the Project Management :

- Project Proposal Writing
- Project planning and scheduling
- Project costing
- Project monitoring and reviews
- Personnel selection and evaluation
- Report writing and presentations

The first four activities are grouped under the project planning, whereas the last two are performed when running the project. All these activities are performed iteratively.

- Writing a proposal is the initial step of a software project. While writing the project proposal it is needed to justify why the particular project is assigned to specific team or an organization. This proposal illustrates the project objectives and the plan to carry out the project. The cost and schedule estimation is also contained in the proposal.
- There may be various organizations competing to get specific project contract. And among several proposals only more effective proposal is accepted and awarded the contract to the respective organization. Many of the organizations' existence are depending on the number of proposals it have. That's why the proposal writing is a critical task. The proposal writing skill is achieved by experience.
- Project planning involves first identification of the activities, milestones and project deliverables, and then a plan is formed which is to be executed for developing the project to produce estimated product in time and with good quality.
- Cost estimation involves the estimation of resources which are needed to successfully carry out the project plan.
- An ongoing project activity, i.e. Project monitoring is done by project manager where he / she keeps examining the project progress and compare it with planned progress and costs. Many organizations use formal methods to do project monitoring. But also by informal discussion with project staff, a manager can understand the project progress.
- The Informal monitoring predicts the possible risks, for example, because of daily discussions with project staff the manager will know that :
 - o The project budget is low so instead of highly paid staff, less experienced and less well-paid staff should be assign to this project.



- The appropriate experienced staff is assign to other important project so there is a need to hire new staff to the project.
- The inexperienced staffs are developing their skills while working on that project and at least on project member is experienced and familiar with the system used to develop the project. If no one is experienced there may be chances of many simple mistakes made by the project members.
- The client and the organization who wins the contract get the report of the project by project manager. The Project manager also responsible to write short and abstract critical information from detailed project report and use this information during the progress reviews.

Responsibilities of a Project Manager :

A person who is responsible for the software project management is known as software project manager. He / She controls and monitors the project management activities performed while developing the software product. He / she is responsible to do following tasks :

The project manager manages the people :

- The project manager acts as project leader
- The project manager Liaison with stakeholders
- The project manager manages the human resources.
- The project manager is responsible to set report hierarchy.

The project manager manages the project :

- The project manager defines and sets project scope
- The project manager handles the project management activities
- The project manager monitors the performance and progress of the project
- The project manager analyzes the risk at each phase of development and handles them if exists.
- The project manager also acts as project spokes person

1.5 Project Planning

- Before starting the actual production the planning is done. It is an ongoing project management activity which starts form initial stage of the project development and continues till the project delivery.
- It involves creation of the set of plans which will help project members to guide and manage time, cost, quality, risks.
- Plans are regularly revised so as to avail the new updates.

Project planning involves following processes :

- **Scoping** : In project planning initial process is to define the project scope, and determining the appropriate methods to accomplish the project. In this in-scope requirements are specified which facilitates the work-breakdown structure creation.
- **Preparing the work breakdown structure** : Followed by scope planning work breakdown structure is constructed. In this the project is spitted into tasks and subtasks, and forms a work breakdown structure by grouping and listing the required duration of various task to accomplish the work.
- **Project scheduling** : This will list the schedules of activities involved in the project and the implementation sequences.



- **Resource planning** : It includes the things which are responsible to accomplish the project. That is it will signify who will do what work? At which time? And if which special skills are needed to accomplish the tasks.
- **Budget planning** : It will estimate how much cost or budget is required to accomplish the project. Also the required resources and the ways in which they are used to balance the cost and time are estimated.
- **Risk management** : It involves the risk planning and migration strategy used to handle the risks.
- **Quality planning** : To generate the quality product. The quality of product is checked with respect to some already defined quality criteria.
- **Communication planning** : This will plan the communication strategy used to communicate with stakeholders.
- **Directing** : Project manager directs the different activities of his team members. He co-ordinates, delegates, motivates, advises, appraises and rewards his team members.
- **Controlling** : The project manager in order to avoid problems and delays, must continuously monitor and report progress against goals, schedule and costs and make appropriate adjustments whenever necessary.
- **Closing** : The last stage involves the project manager in assessing the project success and failure. He learns from the mistakes made and plans for continuous improvement of the project.

Need of software project management plan :

- It is a major part of project agreement.
- It controls the software development process.
- It describes the technical and managerial concepts involved in software development.
- It is a support for requirements analysis document.

Project plan is a document that defines :

- The scope (possible functions and limitations of the system), duration (time required to develop the system), cost (estimated budget) and deliverables for the project.
- System delivery dates, delivery location. Deliverables may be :
 - Documents – user manuals
 - Function descriptions
 - Non-functional requirement descriptions
 - Components descriptions

1.5.1 Structure of a Software Project Management Plan

- Front matter
- Title page
- Revision sheet (update history)
- Preface: Scope and purpose
- Tables of contents, figures, tables.

1. Introduction :

- (i) **Project overview** : It involves summarized description of project and product.



- (ii) **Project deliverables** : It involves the description of items to be delivered, their delivery dates and delivery location.
- (iii) **Evolution of the SPMP (Software Project Management Plan)** : It involves descriptions of anticipated and unanticipated changes.
- (iv) **References** : It includes complete list of materials referenced in SPMP.
- (v) **Keywords, Definitions and Acronyms.**

2. Project organization :

- (i) **Process model** : Describes inter-relationships among various project elements regarding functions, activities, tasks, Milestones, Baselines, Reviews, Work breakdown structure, Project deliverables and Sign-offs.
- (ii) **Organizational structure** : Describes internal management and organizational chart.
- (iii) **Organizational interfaces** : Describes relations with other entities.
- (iv) **Responsibilities** : Describes the major functions and activities.
States responsibility of each team member as who is in-charge of which activity.

3. Managerial process :

- (i) **Management objectives and priorities** : Philosophy, goals and priorities.
- (ii) **Assumptions, dependencies, constraints** :
 - Assumption may be that the security will not be addressed.
 - Dependency may be that the automatic code generation facility in the CASE tool depends on JDK.
 - Constraints may be that the length of the project is 3 months. Limited amount of time to build the system or the project consists of beginners and so, it will take time to learn how to use the tools.

- (iii) **Risk management** :

Identifying, assessing, tracking Contingencies for risks

Example :

Risk : Members in key roles drop the course.

Contingency : Roles are assigned to somebody else. Functionality of the system is renegotiated with the client.

Example :

Risk : The project is falling behind schedule.

Contingency : Extra project meetings are scheduled.

- (iv) **Monitoring and controlling** : It involves the reporting mechanism of information flows and code reviews.
- (v) **Staffing Plan** : It describes the required skills needed for project management.

4. Technical process :

- (i) **Methods, Tools and Techniques** : Computing system, development method, team structure, Standards, guidelines, policies.
- (ii) **Software documentation** : Documentation plan, including milestones, reviews and baselines.
- (iii) **Project support functions** : Plans for functions (quality assurance, configuration management).

5. Work Elements, Schedule, Budget :

- (i) **Work packages (Work breakdown structure)** : Project decomposed into tasks; definitions of tasks.
- (ii) **Dependencies** : Defines relations among the functions, activities and tasks in a priority.
- (iii) **Resource requirements** : Lists the estimations of the resources such as development staff (personnel), development time, special hardware and support software, if any.
 - (a) **Budget and resource allocation** : Defines the costs related with functions, activities and tasks.
 - (b) **Schedule** : States the deadline, dependencies and the project milestones.

1.5.2 Project Communication and Documentation

Project Communication :

- A Communication plan is an important document of the project.
- Communication and documentation is a base for a successful project.

This communication plan includes ;

- Analysis of the stakeholders - Whom to notify if something goes wrong?, How will the issues be communicated? and When to communicate ?
- Background for the plan
- Clearly stated communication objectives

Need of communication documentation :

- **It is a proof of performance** : If the communication plan is not documented, that means, it did not happen. And if you have documented it, that means, you agreed to do it.
- It is an agreement on how to proceed.
- It tracks the progress of the project.
- It provides a functioning interface between units
- Helps in "end-to-end" thinking
- It is an agreement on roles, tasks, schedule

Project Documentation :

Various types of documents that are maintained while project management are ;

Project status report

- It covers the list of tasks and states the specific period of time in which it should get completed.
- It describes the completed as well as the tasks that were incomplete when they were scheduled.
- It lists the upcoming tasks, any issues that they have come across in the duration of the project, changes (if any) to the project schedule.
- It communicates the overall status of the project to all the team members.

Change management plan

- It describes what and how the change will be enforced in the course of the project.

- For example, if we are using a project management tool and sometime later, if another tool becomes available that is more user friendly than the one that is being used, then change management plan replaces the current tool with the new one, thus, ensuring that the transition runs more smoothly.

Risk management plan

It includes anything that might go wrong in the various stages of the project development such as the risks associated with project planning process.

Closing reports

- This report is created to ensure that the project has been completed.
- It communicates to the stakeholders about what the project consists of.
- The stakeholders can look back at the project and see what succeeded and what didn't work.
- It acts as a valuable resource for the future when project managers take over a similar kind of project.

1.5.3 Plans, Methods and Methodologies

- **Plan :** It includes some planned phases and a method or a procedure of work.
 - o Elicit the requirements
 - o Analyze the requirements
 - o Design the test cases
 - o Design the model
 - o Build the code
 - o Test the code
 - o Compare actual and expected results
 - o Install
 - o Maintain
- **Methods :** It is an activity defined in the plan. The plan converts the method into real activities so as to achieve a goal. The method includes :
 - o Activity start and end date
 - o Who is responsible to carry out that activity
 - o What tools and resources are required to perform that activity
- **Methodologies :** The output of one method might be the input to perform the next method. Group of all such methods or techniques forms a methodology such as an OOA (Object oriented Analysis) or an OOD (object oriented design).

1.5.4 Categorizing Software Projects

Compulsory v/s. Voluntary users

There are some tasks that the users of the system have to compulsorily do such as billing the products and in such case developers can get precise requirements of that task but there are also tasks that are voluntary such as in case of the computer games - while developing gaming software, developers cannot get precise requirements.



1. **Operating system** : The system that boots a digital system.

Examples : Windows OS, Linux OS

2. **Information systems v/s. Embedded Systems**

- **Information system** : The system interfaces with the organization. An inventory management system is an information system that controls the stock when the organization reorders the stock.
- **Embedded system** : The system interfaces with the machine. A process control system might control the CCTV cameras in a building.

3. **Outsourced projects**

While developing a large and complex project, some parts of the project are out sourced to other companies in the cases when the company doesn't have sufficient expertise, or the company understands that some parts can be developed cost-effectively by another company. Such outsourced projects are usually small in size and should be completed within a few months duration.

4. **Objective-driven development v/s. Product-driven development**

- Object driven development focuses on identifying the need for a new software system.
- Product-driven development focuses on producing a product based on the details provided by the client in the specification.

1.5.5 Project Charter

- It states the project objectives, project goals, roles and responsibilities.
- It lists out the reasons for undertaking the project.
- It provides solutions to the problem in hand.
- It identifies the main stakeholders and the authority of a project manager.
- This document is required by Initiative for Policy Dialogue (IPD) and Customer Relationship Management (CRM).

Benefits of Project Charter

1. It improves customer relationships.
2. It improves project management processes.
3. Improves Regional and headquarter communications.
4. Helps in gaining project sponsorship
5. It recognizes senior management roles and authorities.
6. Allows progression, which is aimed at attaining industry best practices.

Elements in Project Charter

Project charter is a project planning tool that address the following key elements :

- Identity of the project.
- Time : Start and end date for the project.
- People involved in the project.
- Project objectives and set targets.



- The Business case.
- Detailed description of a problem
- The ROI expected from the project.
- Performance of the project
- Roles and responsibilities of the participants involved.
- List of resources needed to achieve the objectives
- Barriers and the risks involved with the project.
- An effective communication plan.

1.5.6 Stakeholders

- These are the people who are interested in the project.
- Stakeholders can be categorized as :
 - o Internal to the project team : Project managers, technical managers, team leaders and developers.
 - o External to the project team but within the same organization : Company owner, share holders and other employees.
 - o External to both the project team and the organization : Government, military, customers, third-party vendors, competitors.

1.5.7 Setting Objectives

- Stakeholders are responsible for setting the objectives of the project.
- The objectives define what the project team must achieve for project success.
- The objective for software developers could be to keep the development costs within a certain budget.
- The objectives should be SMART i.e. specific, measurable, achievable, relevancy and time constrained.

1.5.8 Business Case

Business case states the reasons to carry out a project charter. It states the benefits that the customer may gain by carrying building a project.

Elements of a good business case are :

- The reasons of undertaking the project.
- The benefits gained from undertaking the project.
- The consequences of not building the project.
- The factors that conclude that the proposed project fits the business goals.
 - o Development costs should be within a certain budget. It should not be increased.
 - o Features of the system should not be reduced without consulting the customer (client)
 - o Delivery date of the project should not be delayed.

1.5.9 Project Success and Failure

Factors that make a project successful :

- On time delivery
- Set of completed features
- Reliable
- Performance
- Meets expectations
- On budget
- Maintainable and enhanceable
- Around 80-85% of software fail.

Reasons for software project failure :

- Insufficient planning: lack of a structured plan
- Unrealistic expectations
- Insufficient risk management
- Poor communication between the stakeholders (managers, developers, testers, marketing, sales, customers)
- Requirement creep : Frequent change in the requirements
- Ambiguous requirement specification : Misunderstanding of requirements
- Gold plating of the requirements
- Poor understanding of the goals
- Complexity of the product
- Switching the tools in middle of the project
- Changes in the budget and priorities
- Doing something without customer's feedback
- Market competition

1.6 Four P's of Project Management

The Four 'P's of Effective Project Management :

1. **People** : People are involved in the process of Recruitment, Selection, Performance management, Training, Organization, work Design, and team/culture development.

There are 5 different types of people who are involved in project management :

1. **Senior manager** : Defines the business issues of the project.
2. **Project manager** : Involves in planning, motivating, organizing and controlling the practitioners.
3. **Practitioners** : Deliver the technical skills in building a product.
4. **Customers** : Specifies the software requirements for the product being developed.
5. **End-users** : Interacts with the developed software product.

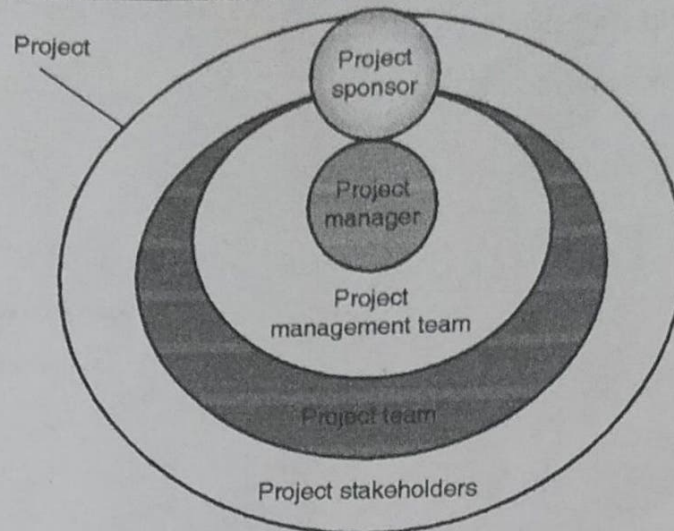


Fig. 1.6.1 : Relationship between stakeholders and project

2. **Product :** To develop a project plan, the problem to be solved should be decomposed. A project scope should be clear, unambiguous and understandable at the management and technical levels. The planning of a software project begins with :
 - (a) Establishing product objectives and scope.
 - (b) Identify technical and management constraints.
 - (c) Using the above information, reasonable cost, project schedule and risk involved can be defined.
3. **Process :** To develop the software, it is necessary to select an appropriate process model or combination of the models among the different process models available i.e. Waterfall model, iterative waterfall model, prototyping model, RAD model, and spiral model.
4. **Project :** A project is a temporary endeavor undertaken to produce a unique product or service. A project is a unique process, consisting of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective that conforms to specific requirements, including the constraints of time, cost, and resources. Projects differ from operations, such as manufacturing, in that operations are ongoing and repetitive, while projects are temporary and unique. A successful project is a step-by-step sequence followed where accurate decisions are to be taken. Managing Successful software projects needs to prevent and avoid the risks/problems involved.

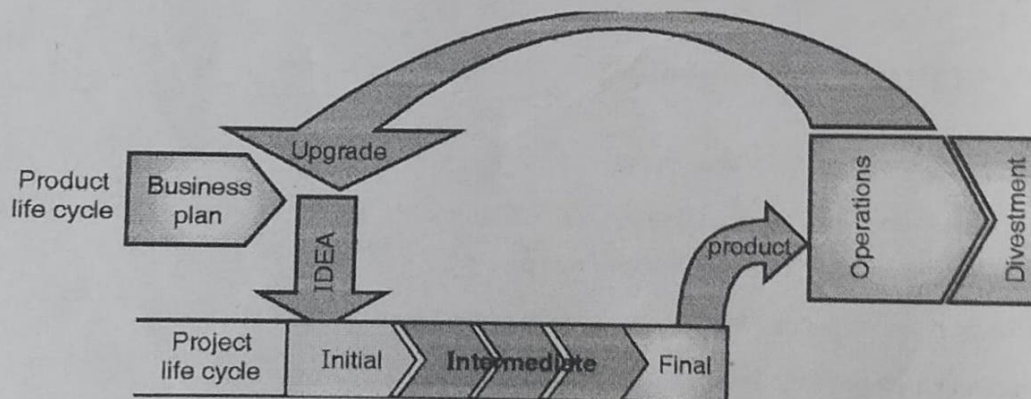


Fig. 1.6.2 : Relationship between product and project life cycle

1.7 Management Control and Knowledge Areas

- Management control is about setting the performance standards and continuously evaluating the performance of the project against these predetermined standards.



- This comparison is done in order to ensure that the performance is in-line with the standards set otherwise, take remedial actions to improve the performance.
- Knowledge areas describe the key elements that a project manager must focus on.

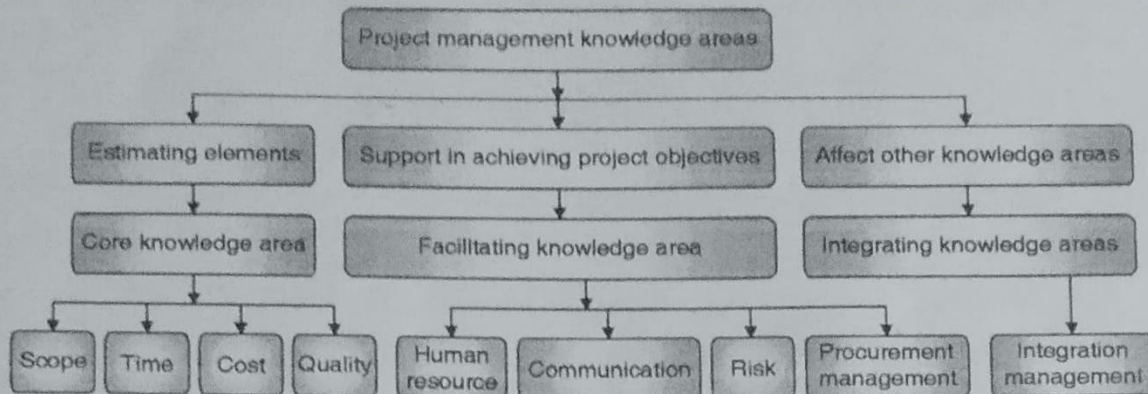


Fig. 1.7.1 : Project management knowledge areas

1. Scope management :

- It describes what the project is supposed to accomplish and the estimation of time and money that is available to achieve these objectives.
- It describes how requirements, details and processes are managed.
- Changes to the scope should be handled in a structured, procedural, and controlled manner. Any change to the scope of the project has a matching change in estimation.

Example : If the project scope is to train 100 users with a budget of Rs. 50, 000 the project manager is expected to do that. But, if the scope is changed to train 150 users, the project manager must do the appropriate changes in the budget estimation. If the budget is not adjusted, then the project manager will avoid the change in scope.

- The project scope must be communicated very clearly to all stakeholders.

2. Time management :

- This process ensures that the project is completed in estimated schedule.
- This process manages the resources, activities, schedule management.
- It defines and sequences the activities; estimates the duration and resources needed for accomplishing each activity.
- Initially schedule is referred to as the time **baseline** of the project. It is later used to compare updated baselines with the original baseline.
- Special softwares are used to build and maintain the schedule and baselines.

3. Cost management :

- This process manages the cost and overall budget estimation. After the cost of the project is estimated, the project manager controls the cost and makes required changes to the budget.
- The Project Cost depends on the cost estimated for each activity in the project. The estimation often changes as the project progresses.

Example : Initially project the cost is more difficult. Later in the project when the scope and the schedule has been defined in detail manner it is easy to find.



- Every year the project manager develops a full and comprehensive budget for the project for three annual periods. Later that budget is approved by donors, the project committee and trustees.
- The project manager must allocate monthly expenses so that the financial statements can be processed every month.
- The project manager must submit a monthly report to the committee chairperson that gives a comparison the annual budget to the monthly expenses.
- The project manager then reports to the team members about the agreed balanced scorecard format.

4. Quality management :

- This process ensures quality control and assurance by focusing on the project meets its requirements i.e. it does what it is intended to do. The quality control ensures that the outputs conform to the quality standard; whereas, the quality assurance focuses on the quality process improvement.
- This process measures the outputs of different processes against some predetermined acceptable measure.
- The project manager creates a quality management plan early in the project development because decisions made about quality can have a significant impact on other decisions about scope, time, cost and risk.

5. Human resource management :

- This process focuses on HR planning like roles and responsibilities, project organization, and staff management.
- It involves assigning staff, assessing and managing the performance of project team members.
- Project manager manages the persons of Business Support Project, project committees, beneficiary committees, trustees, donors and volunteers.

6. Communication management :

- This process decides what information is needed, how that information will be sent and managed, and how project performance must be reported.
- This process focuses on properly communicating the project's stakeholders about the entire project.
- Communication is a proactive and thorough mixture of formal and informal, written and verbal information.
- The project manager communicates the project information accurately in a timely manner to all the stakeholders.
- This process focuses on creating a communications plan which describes the kind of information (for e.g. The project performance) that should be communicated on a regular basis to all those who should receive it.

7. Risk management :

- This process designs a plan describing how to handle risks in the project - identifying risks, and how to respond to these identified risks.
- Risks arise due to uncertain events and may affect the project for good or for bad.
- Assessing of risks includes - how likely will the risks happen, how it will affect the project, and what will be the cost if it arises ?
- The project manager uses risk analysis tools and techniques to assess the risks and answer the questions.

8. Procurement management :

- It focuses on how to obtain goods or services from an outside organization.
- The project manager plans for purchasing and acquisition of products and services that can't be provided by own organization.
- The plan includes preparation of procurement documents, request to vendor responses, selection of the vendors and creating and administering contracts with each vendor.

9. Integration management :

- This process coordinates the other 8 knowledge management areas to work together throughout the project.
- It focuses on the big tasks that must be done for the project to work. Every part of the project is coordinated.
- When the project starts, the project plan is assembled and executed, the work is monitored and verification of the results of the work is performed, and also performs the tasks associated with closing the project.
- This area involves the management and control of the entire project.

1.8 Project Management Life Cycle

- The project life cycle describes the tasks that must be completed to produce a product or service.
- The project manager needs to be aware of how the inputs and outputs of one lifecycle affect and shape them.

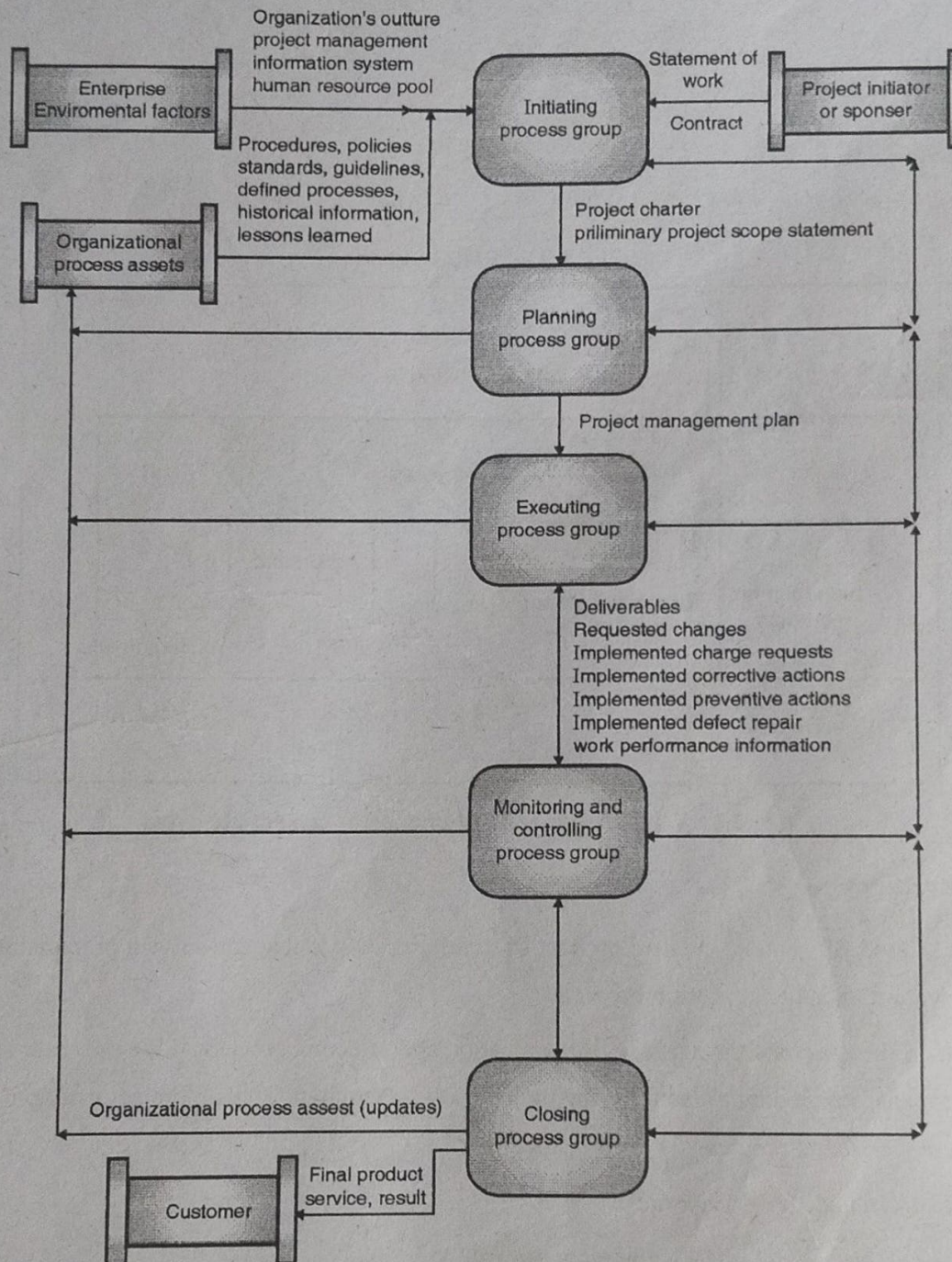


Fig. 1.8.1 : Project management life cycle

As no two projects are exactly alike, all projects should progress through different phases.

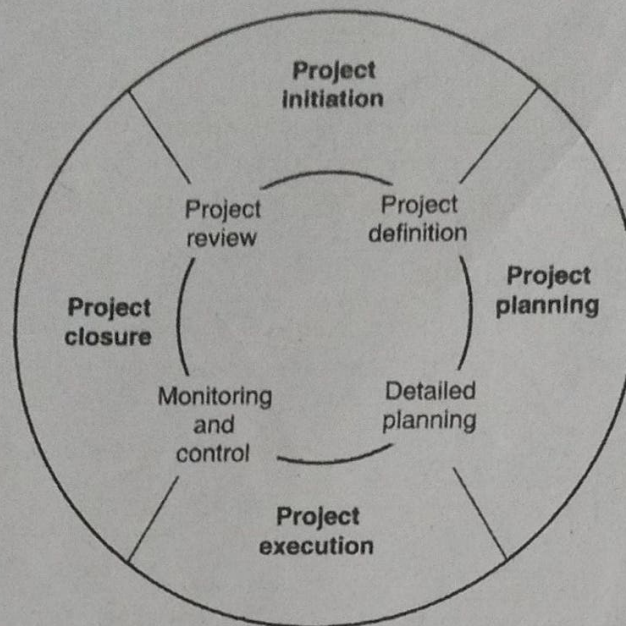


Fig. 1.8.2 : PMLC phases

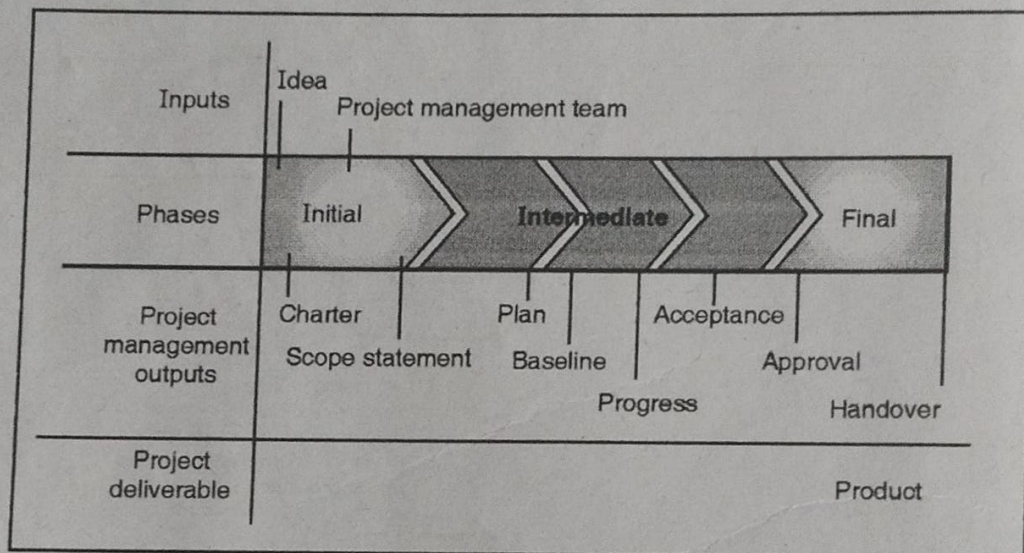


Fig. 1.8.3 : Typical sequence of phases in project life cycle

1. Propose the project :

- When an individual wants to create a product or develop a service that can solve a problem or address a need of his organization, then he proposes for a project.
- Organization then submits the proposal for evaluation and selection process. If the proposal is selected, a budget or further management commitment for the project may be required. Then, a project manager is assigned and the project is authorized to progress for project initiation.
- Tasks performed under this phase are :
 - (i) Develop business case, Develop proposed solution.
 - (ii) The proposed solution will be input to the next phase i.e. Project Initiation.

2. Project initiation :

- Initially a project manager is assigned. Then the project manager works with project sponsor to identify necessary resources and team members. Then they find the key project parameters – cost, scope, schedule and quality (CSSQ). The project team prepares a document called project proposal, which includes initial business case. If the project sponsor approves this, then the designated team begins the initial planning effort. Project Plan also involves and communicates with all the parties to identify foreseeable risks that can threaten the project.
- At the end of this phase, business case is revised and re-evaluated and a decision is made to either halt the project or proceed to project planning. The output of this phase will be approved project proposal, which is input to the next phase.
- **Tasks performed under this phase are :**
 - (i) Identify sponsor.
 - (ii) Identify project team.
 - (iii) Review historical information.
 - (iv) Develop project proposal.
 - (v) Conduct kickoff meeting.
 - (vi) Establish project repository.

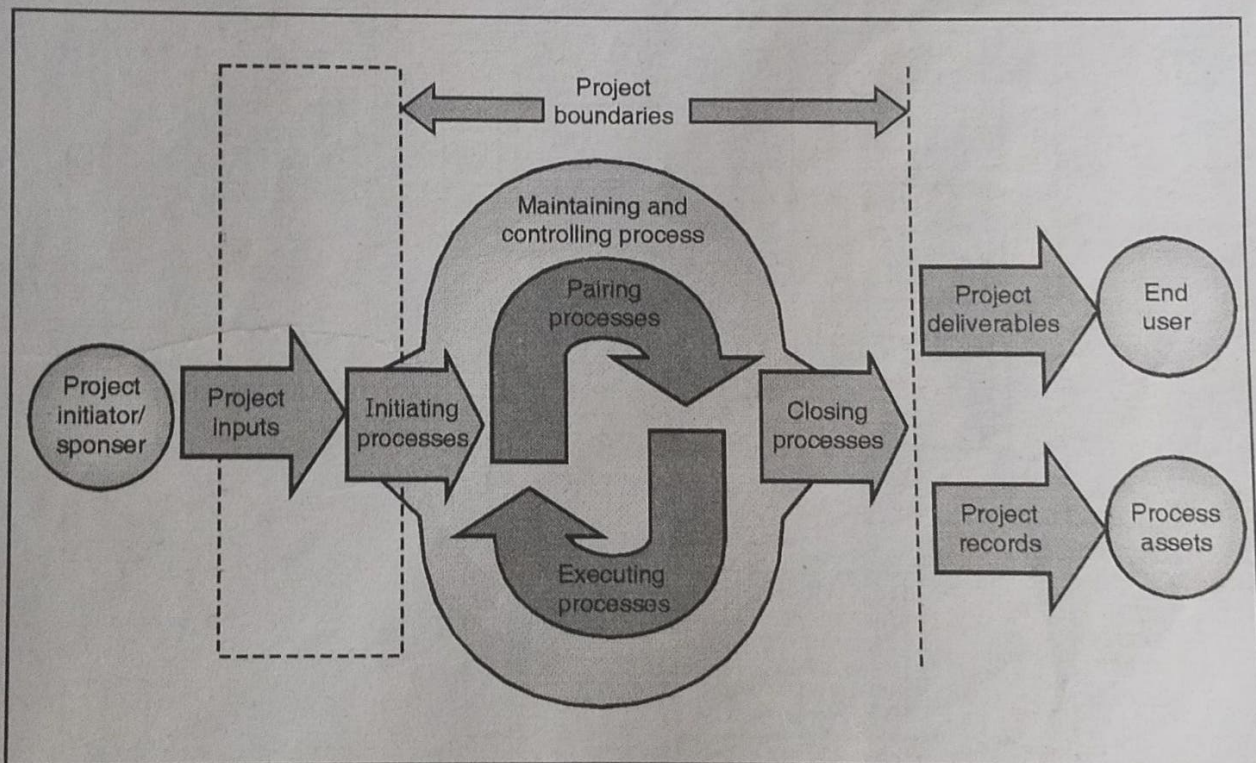


Fig. 1.8.4 : Initiating project boundaries

3. Project planning :

- It conducts kickoff planning. Project planning builds on work done in project initiation, refining and improving Cost, Scope, Schedule and Quality and project deliverables.

- Mostly extra members join project team and they assist project manager in further elaborating CSSQ. Other key elements like change control, acceptance management, issue management, organization change management and project transition are added to project plan including project specification items.
- Project planning marks completion of project plan. Lastly business case is revised and re-evaluated based on completed planning documents. Finally decision is made again either to halt the project or to proceed further for project execution and control.

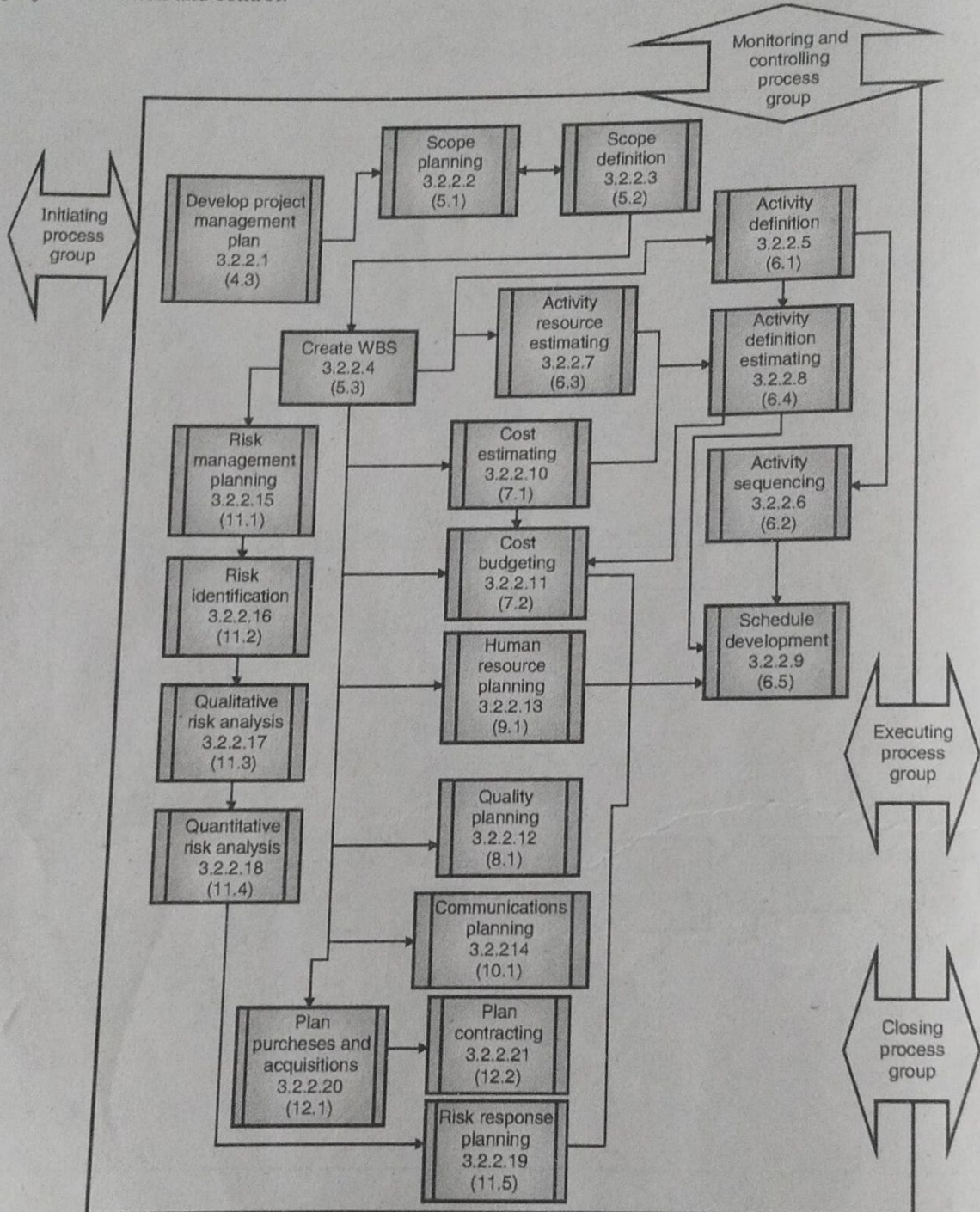


Fig. 1.8.5 : Planning process group

Tasks performed under this phase are :

- (i) Orient new team members.
- (ii) Review project material.
- (iii) Kickoff project planning.

4. Project execution and control :

Conducts phase kickoff. Most of the resources are applied/expanded on the project. Initially lots of team members are recruited into the project. The primary task of project manager is to allot the team members to execute the tasks on defined project schedule and develop the product or service the project is expected to deliver. Project manager uses processes and plans prepared during earlier phases.

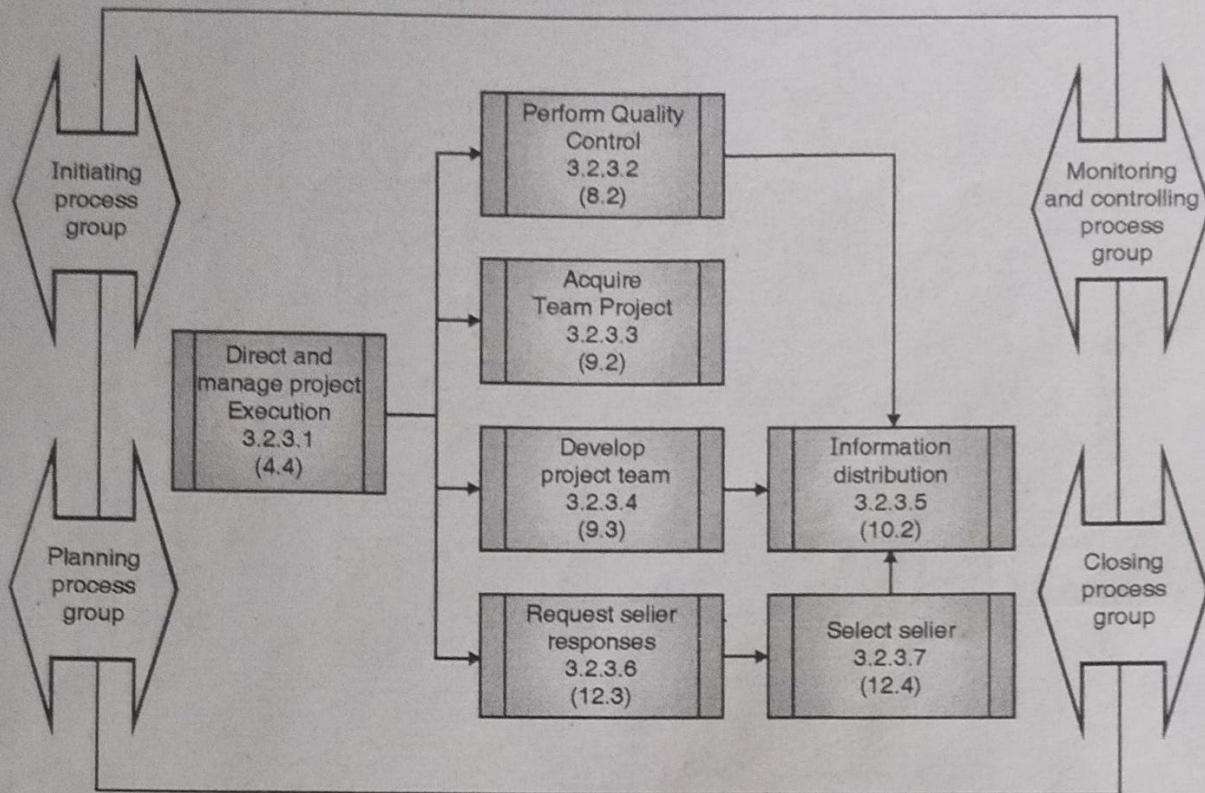


Fig. 1.8.6 : Executing process group

Tasks performed under this phase are :

- (i) Orient new team members.
- (ii) Review project materials.
- (iii) Kickoff project execution.

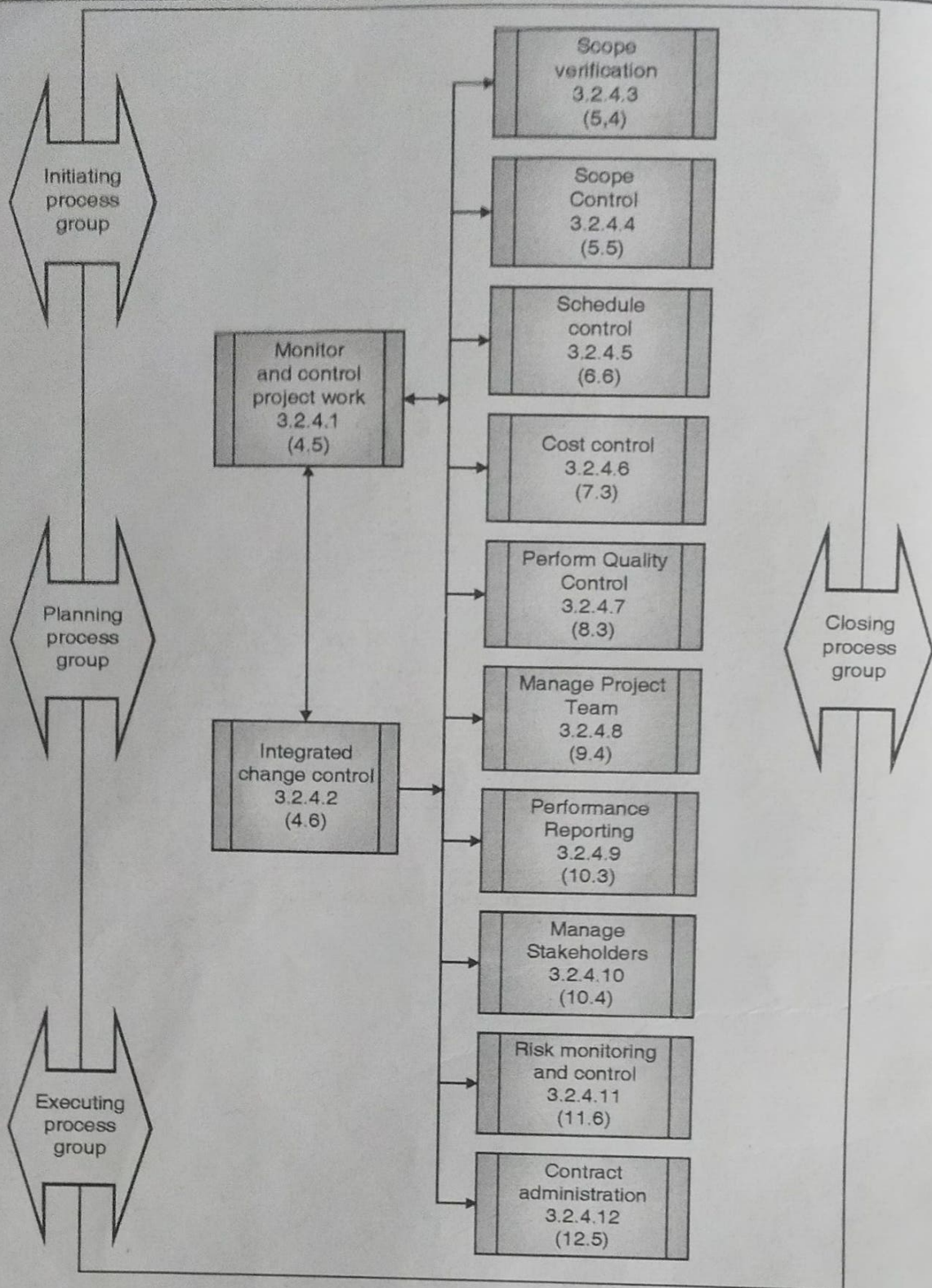


Fig. 1.8.7 : Monitoring and controlling process group

5. Project closure :

Conducts post implementation review. The Project team assesses the outcome of project as well as performance of team and the organization. This is done by feedback evaluation from customers, team members, consumers and other stakeholders. The Review documents the best practices and these lessons can be used for future projects. Project metrics captured are used to compare and evaluate performance measurements.

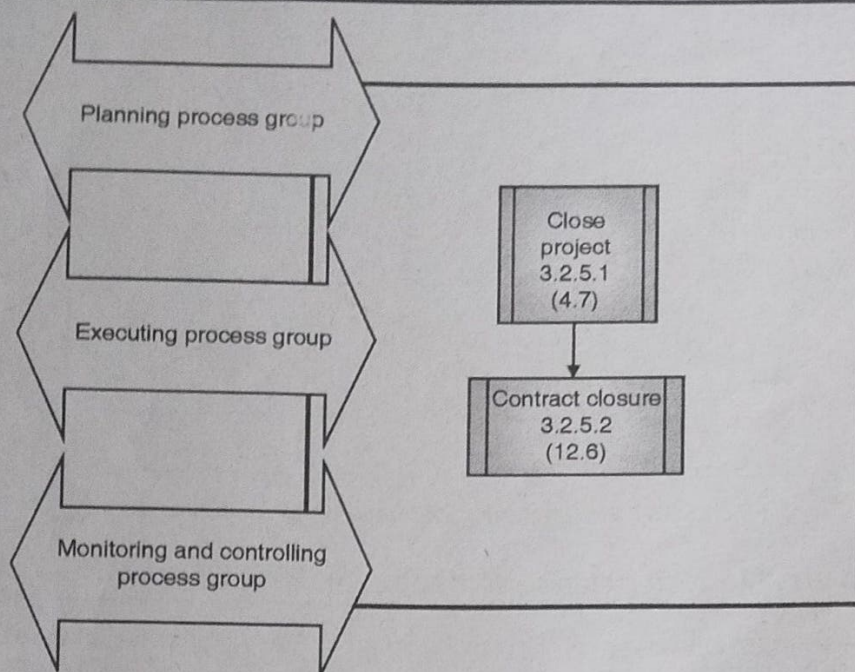


Fig. 1.8.8 : Closing process group

Tasks performed under this phase are :

- (i) Solicit feedback.
- (ii) Conduct project assessment.
- (iii) Prepare post implementation report.

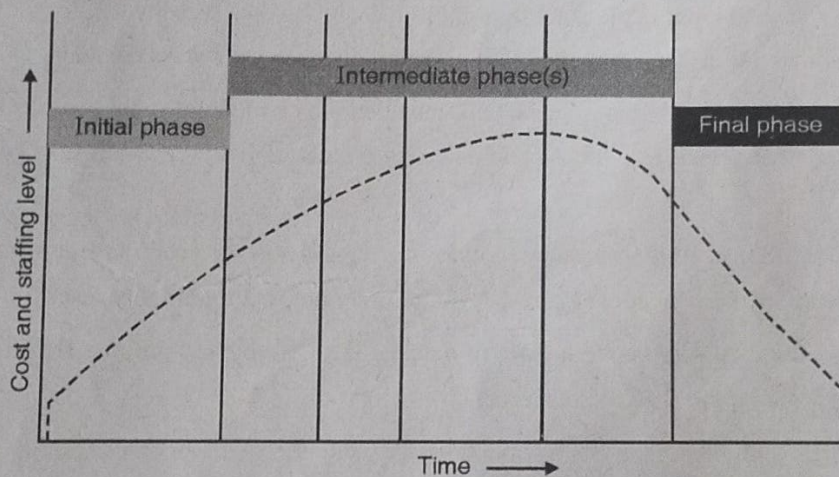


Fig. 1.8.9 : Typical project cost and staffing level in project management life cycle

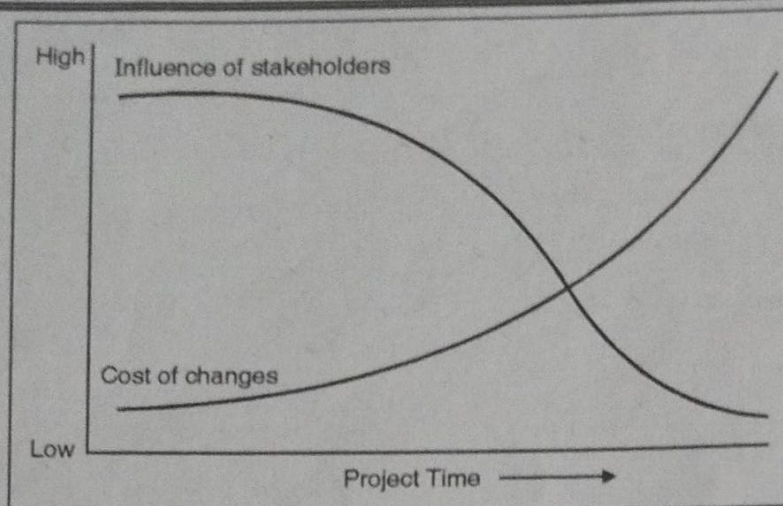


Fig. 1.8.10 : Stakeholder's influence on project over time

1.9 Traditional v/s. Modern Project Management Practices

Regardless of the type of the project, size of the activities or effort, every project requires project management.

There are two main types of project management :

- Traditional Project management is a set of orthodox methods and techniques that can be applied to an activity seeking an end product, outcome or a service'
- Modern Project management Agile and Scrum are two such modern project management methods.

Example :

- Consider a company where in each of its staff uses the same old OS i.e. Windows XP. And now, The management decides on upgrading the OS to enhance the productivity and reduce the OS security threats.
- If you have one office with about 100 computers, it is considered as a medium scale project. In case if your company has 10-15 branches, then the project is a large scale one with high complexity. In such case, you are over-burdened by the tasks at hand.
- This is where traditional project management comes in. Traditional project management does not require any customizations and also modern project management methods are not required in this case.
- The company can use an existing project manager to manage the OS upgrade project. He will plan the entire project, derive a schedule, and indicate the required resources.
- The cost will be elaborated to the higher management. There will be frequent updates coming from the project manager to all stakeholders.

Multiple Choice Questions

Q. 1 What is project management?

- (a) The planning, organizing, monitoring, and controlling of all aspects of the project in a continuous process in order to achieve its objectives.
- (b) The planning and application of business and financial models used to control all aspects of the project for the purpose of meeting project objectives.



Project Evaluation and Programme Management

Syllabus

Introduction, Business Case, Project Portfolio Management, Evaluation of Individual Projects, Cost-benefit Evaluation Techniques, Risk Evaluation, Program management, Managing the Allocation of Resources within Programmes, Strategic Programme Management, Creating a Programme, Aids to Programme Management, Some Reservations about Programme Management, Benefits Management.

2.1 Introduction

- Success of a project in both technical and business terms is based on strong initial evaluation of the technical, organizational and financial feasibility of the project.
- In Some cases managers are able to clearly justify the commitment to a particular project since they know that the financial benefits will clearly exceed their investments.
- But, in few cases, managers may not be able to approve a project on its own, but can see that it fulfills the strategic objectives when combined with the outcomes of other projects.
- Strategic assessment is where a group of projects are managed and coordinated in a way to gain benefits that would not be otherwise possible if the projects were managed independently.
- Programme management sees the individual projects as components of a programme and evaluates and manages them as collection of projects that all contribute to the same overall organizational goals.

2.2 Business Case

- The business case evaluates the benefit, cost and risk involved in the project or a programme and then provides justification (reasons) to undertake that project or programme.
- It is a good practice to have a business case as it depicts the value of the work done throughout that project.
- Senior management goes through the outline of the business case and then decides to give the go-ahead so that a detailed business case can be prepared.
- Once approved, the business case is kept up to date reflecting all the approved changes, In this way, it is used as a primary document to determine if the work should continue.

A business case includes :

- Strategic case : Overview of the project / programme and why is it needed
- Options appraisal : Alternate options and which option is the most suitable.
- Expected benefits : Benefits that will be achieved from the project and also the unavoidable disbenefits (if any)
- Commercial aspects : Costs, investments and funding arrangements

- Risk : Major risks and their impact on the business case
- Timescales : Time required to deliver the output and achieve the benefits.

Business case for a project :

Content of the business case should reflect the major project requirements and context. For example :

- If an organization is delivering a project to the client, that organization's business case will be based on the profit gained by delivering the project output and the client's business case will be based on the benefits derived from the output.
- If a project is stand-alone, the project management team will have to develop their own business case; if a project is part of a programme then the business case is provided by the programme management team;
- If the project is large, risks involved are also large. Therefore, the sponsor and other members of senior management require a detailed business case to gain confidence about their work.

Business case for a programme :

- A programme business case must reveal the uncertainty in the programme environment and provide justification based on quantifiable benefits about why to undertake this programme.
- Programmes are usually broken down into branches and each tranche has a separate business case. So if a particular tranche is completed then it must have already delivered some beneficial change.
- Programme management team must monitor the interdependencies between projects so that they can identify if problems occurring in one project are affecting the business case of another and if so, it should be communicated in time.

Business case for a portfolio :

- Since a portfolio is not a stand-alone enterprise but rather a collection of programmes, projects and business, therefore, it does not require its own business case. Each programme and project within the portfolio has its own business case.
- The portfolio management team assesses the business case to categorise, prioritise and balance the portfolio.

2.3 Project Portfolio Management

Both Project Management (PM) and Project Portfolio Management (PPM) have many similar attributes but they are two different and distinct set of processes.

- **Project management** : Suitable for projects developed by software houses for use in their own organization.
- **Project portfolio management** : Suitable for projects developed by software houses for use in client organizations.

Table 2.3.1 : Project management versus portfolio management

Sr. No.	Project Management (PM)	Project Portfolio Management (PPM)
1.	It is about applying all the knowledge, skills, tools and techniques into the project activities so as to meet project requirements	It is about applying all the knowledge, skills, tools and techniques in a portfolio (collection of projects) so as to meet the needs of an organization's investment strategy
2.	Stakeholders : Business owners, sponsors, end users	Stakeholders : Financial management, senior executives, stockholders

Sr. No.	Project Management (PM)	Project Portfolio Management (PPM)
3.	Stakeholder needs : Business requirements, controlling costs and schedule	Stakeholder needs : Optimal investment of resources, ROI, strategic alignment
4.	PM Process : Initiating, Planning, Executing, Closing, Monitoring and Controlling.	PPM Process : Identifying, Categorizing, Evaluating, Selecting, Prioritizing, Balancing, Authorizing, Performance Reporting, Monitoring and communicating adjustments.

Integrating PM and PPM helps organizations to choose best portfolio of projects that are aligned with organizational strategic goals and performance and also allows to re-prioritize the business needs and change the budget.

PPM Benefits :

1. Ensures business alignment
2. Provides justification for its investment decisions that leads to a right project mix
3. Reduces resource conflicts and constraints
4. Collective management of all projects i.e. known as portfolio management enables identification of tradeoffs
5. Monitoring and controlling i.e. helping, modifying, slowing, and stopping projects when necessary
6. Drives down the project cycle-time if necessary to improve the project performance.

2.4 Evaluation of Individual Projects

Need of project evaluation :

Every project needs an high level evaluation (assessment) :

- To check if it is worth to proceed with the project
- To check that it is not too late to proceed with the project
- To check if the project fits in the strategic planning of the whole organization
- To identify the most suitable project after analysing the available alternate projects so as to achieve a better success rate and a higher turnover
- To check if it is desirable to carry out the development and operation of the software system.

Who does the project evaluation :

- Senior management
- Project manager/coordinator
- Team leader

Project evaluation is usually done at the beginning of the project. For example, it is done at Step 0 of the Step-wise framework.

Categories of project evaluation (assessment) :

- Strategic assessment
- Technical assessment
- Economic assessment

Project evaluation includes :

- Cost-benefit analysis
- Cash flow forecasting
- Cost-benefit evaluation techniques
- Risk analysis

2.4.1 Cost Benefit Analysis

Cost Benefit Analysis (CBA) is a standard way to assess the economic benefits.

Two steps to calculate CBA :

1. Identify and estimate all the costs (development costs, setup costs, operational costs) and benefits (direct benefits, assessable indirect benefits, intangible benefits) of carrying out the project.

Costs :

- Development costs : Salaries, employment costs of staff
- Setup costs : Cost of equipment, hardware, database conversion, staff training
- Operational costs : It consist of the 4:0% of operating the system once it has been installed.

Benefits :

- Direct benefits : Benefit gained by reducing the staff and re-organizing the staff
 - Assessable indirect benefit : These are secondary benefits such as increase of accuracy through a more user-friendly screen
 - Intangible benefits : These benefits are longer term in nature such as enhanced job interest leads to reduction in recruitment costs.
2. Express the costs and benefits in a common unit for easy comparison (e.g. \$).

2.4.2 Cash Flow Forecasting

- Cash flow forecasting is the estimation of the cash flow over time.
- Cash flow forecasting is required if in case an excess of estimated benefits over the estimated costs is not sufficient or if we need detailed estimation of benefits and costs versus time.
- Forecasting is not very easy since much information is not available in early phases of the project and also the project may span for several years.
- Need to revise the cash flow forecast every quarterly or even monthly.
- It depicts both **expenditure** and **income** :
 - o Expenditure : Staff salary, recruitment costs, bank interest. If the money is from bank, you need to calculate the interest as well.
 - o Income : Payment on completion or Payment by phases (payment in phases occurs in outsourcing projects)

2.4.3 Cost Benefit Evaluation Techniques

- Cost benefit evaluation process identifies :
 - o All the costs such as Development costs, Set-up cost, Operational costs
 - o Identify the value of benefits
 - o Checks if benefits > costs



Term to know while evaluating cost benefit :

- Cash-flow is the value of income-less outgoing i.e. calculated at the end of each year.
- Net profit is the value of all cash-flows for the lifetime of the application
- Pay-back period is the time it takes to start generating surplus income > outgoings.

Consider an Example :

Year	Project 1
0	- 100,000
1	5,000
2	10,000
3	10,000
4	20,000
5	90,000
Net profit	35,000
Payback	5
ROI	7%

- Year 0 represents initial investment made at the beginning of the project i.e. all the costs before system is in operation
- Negative values represent net expenses
- Positive values represent net incomes

Table 2.4.1 : Cost benefit evaluation techniques

Techniques	Formula	Advantage	Disadvantage
Net Profit	$= \text{Total Income} - \text{Total Costs}$ In the above example; Total Income $= 5000 + 10000 + 10000 + 20000 + 90000 = 135000$ Net Profit = $135000 - 100000 = 35000$	Simple to use	Ignores the timing of the cash flow
Payback period	Time taken to break even In the above example; Payback Period = 5	Simple to calculate, not very sensitive to small forecasting errors	Ignores any income or expense after the payback period
Return on Investment	$\text{ROI} = \frac{\text{Average annual profit}}{\text{Total investment}} \times 100$ In the above example; Avg. annual profit = $45,000/5 = 7,000$ ROI = $(7,000/100,000) \times 100 = 7\%$	Simple and easy to calculate quite popular	- Ignores the timing of the cash flow - Potentially it is very misleading because it is very tempting to compare the rate of return with the current interest rates

Techniques	Formula	Advantage	Disadvantage
Net present value	NPV is the sum of the present values (discounted cash flows) for all the years of the project. Discount rate(r) is the annual rate by which we discount future earnings. For example, If $r = 10\%$ and $ROI = \$110$ in a year, then the Present value of the investment is \$100 for that year. $PV = \frac{\text{Value in year } n}{(1 + r)^n}$ $PV = 110/(1+0.10)^1 = 100$	Considers the profitability of a project and the timing of the cash flows that are produced.	- Difficult to select an appropriate discount rate - NPV might not be directly comparable with earnings from other investments or the costs of borrowing capital.
Internal rate of return	It is the percentage discount rate that would produce an NPV of zero for the project.	- Can be used to compare different investment opportunities - directly comparable with rate of return on other projects and with interest rates - There is a Microsoft Excel function which can be used to calculate IRR. - It is also supported by Lotus 1-2-3. - It is useful and convenient since further calculations are not required.	- Does not consider important factors like project size, project duration, future costs - It simply compares the project cash flow.

2.4.4 Risk Evaluation

IMP

- Risk is a factor that causes a project to be delayed or over-budget.
- During project evaluation, we make some estimations based on some **assumptions**. Based on these estimations, we decide whether to continue the project or not. Thus, anything that goes wrong with the estimations are potential threats/risks to the project.

Example : Project A might appear to give better ROI compared to project B but could be riskier.

Draw a project risk matrix for each project to assess their risks. In the matrix, **Importance** relates to the cost of damage if the **Risk** has the **likelihood** (probability) to occur. H means High, M means Medium and L means Low.

Risk	Importance	Likelihood
Software never completed and delivered	H	
Delay in software delivery	M	M
Cancelled the Project after Design stage	H	
Development Budget exceeded < 30%	L	M
Development Budget exceeded > 30%	M	L
Actual Maintenance costs > estimation	M	M

2.5 Programme Management

Definition : A group of projects that are managed in a coordinated way to gain benefits that would not be possible were the projects to be managed independently' – Ferns.

Programme management is suitable for projects developed for use in the organization. Programme management sees the individual projects as components of a programme and evaluates and manages them as collection of projects that contribute to the same overall organizational goals.

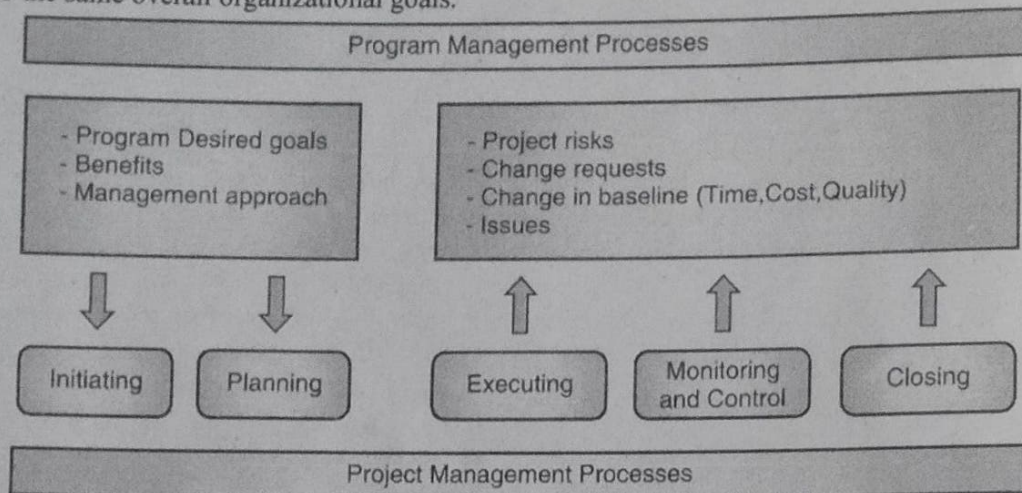


Fig. 2.5.1 : Programme and project management

Example :

Customers' experience of the organization can be very variable and inconsistent. The employee who records the customer's requirements is different from the people who actually carry out the work and different again from the clerk who deals with accounts. Different maintenance engineers deal with different types of equipment. A business objective might be to present a consistent and uniform interface to the client. This objective might need changes to a number of different systems which until now have been largely self-contained. The work to reorganize each individual area might be treated as separate projects, co-ordinated at a higher level as a programme.

- Effective programme management requires a well-defined programme goal and then all the projects of the organization are selected and tuned so as to contribute to this goal.
- In the programme management, a project is evaluated according to how it contributes to this programme goal and how its viability, timing, resourcing and final worth is affected by the programme as a whole.

Types of Programmes :

- **Strategic :** A group of projects implementing the same strategy are managed and coordinated together in a way to gain benefits that would not be otherwise possible if the projects were managed independently.
- **Business cycle programmes :** A portfolio of project that take place within a certain time period e.g. till next financial year.
- **Infrastructure programmes :** Different ICT-based applications which share the same hardware/software infrastructure in an organization.
- **Research and development programmes :** Developing new products in an innovative environment, some of which are very abstract and high-risk but potentially but very profitable and some will have a lower risk but will return lower profit. Getting the right balance would be key to the organization's long term success.
- **Innovative partnerships :** Developing new technologies that can be exploited by a whole range of companies.

Table 2.5.1 : Programme manager vs. project manager

Programme Management	Project Management
Focuses on the vision through function and design	Provides the structure and mechanism to enable the vision
Focuses on big picture – fulfilment of a strategy and realization of benefits	Focuses on completing the tasks, deliverables and meeting the SRS on time within budget
Group of projects in a programme are very much similar	Every project is unique

Programme Managers	Project Managers
Manage many simultaneous project say, 5-6 projects at a time.	Manage only one project at a time
Maintain personal relationship with skilled resources	Maintain impersonal relationship with resources
Optimization of resource usage - concerned with ensuring the best use of staff ensuring that staff has regular work with no period of idleness between the project tasks. e.g. Programme Manager has allocated Resource W in project A and Project B with no period of idleness between the project tasks.	Minimization of demand for resources. Project manager ensure that there is least dependency on a particular resource and therefore assign alternate resources. e.g. Project Manager has allocated resource W, X and Z for the same project B.

Project managers

		Project A	Project B	Project C	Project D
Program management	Resource W	×	×		
	Resource X		×		
	Resource Y			×	×
	Resource Z	×	×		×

Fig. 2.5.2 : Role of programme manager vs. project manager during resource allocation

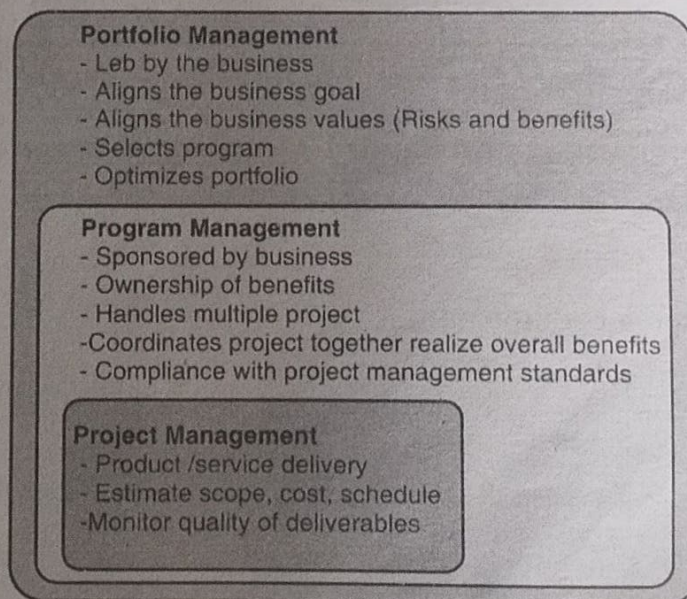


Fig. 2.5.3 : Project, programme, portfolio management

2.5.1 Creating a Programme

Programme life cycle includes following steps :

1. **Pre-programme preparation :** The programme concept is shared, programme is chartered and programme manager identified. This phase includes ;
 - Understanding of strategic benefits of the programme
 - Developing and initiating the plan
 - Defining the programme objectives and aligning them with the organizational goals
 - Developing a business case and understanding the business benefits
2. **Programme initiation :** Generate programme's scope, schedule, activities, costs and other planning outcomes.
3. **Programme setup :** Skeleton of the programme is created that defines how the programme and its projects will operate. This phase includes :
 - Aligning the mission, vision and values of the programme with organizational objectives
 - Developing detailed cost and schedule plan
 - Conducting feasibility studies – technical, economical, ethical feasibility
4. **Delivery of programme benefits :** All the projects under the programme are initiated, the project managers and their teams go about creating the incremental benefits to achieve the goal of the programme. This phase includes following tasks :
 - Establish project monitoring team to monitor and control the projects
 - Initiate projects to meet programme objectives
 - Ensure that the deliverables meet the requirements
 - Assess the progress against the plan
 - Identify environmental changes which may impact the programme management and its anticipated benefits.
5. **Program closure :** This phase includes the following tasks :
 - Review the status of benefits
 - Disband the programme organization
 - Disband the programme team so that it is available to be deployed in other programmes
 - Document the lessons learnt so that they can be used for reference in future – it depicts the weaknesses or areas to improve and best practices to be utilized in future.

2.5.2 Aids to Programme Management

Definition : Program management is the coordinated management of interdependent projects over a finite period of time in order to achieve a set of business goals.

The program manager's role is to keep all projects moving in the same direction to achieve the business goals outlined in the business case.

Programme management is composed of :

- **Strategic formulation** : Pressure to change, corporate objectives, business needs.
- **Value management** : Strategic objectives, expected benefits, expected outcomes
- **Project management** : Proposed actions, proposed deliverables, delivered outputs
- **Operational management** : Enhanced capabilities, realized outcomes. Realized benefits

Programme management is composed of :

- Programme manager
- Change agents
- Quality manager
- Governing body
- Programme sponsor
- Project manager
- Technical authority

2.5.3 Some Reservations about Programme Management

Programme management provides following benefits :

1. Resources, talent and expert judgement are shared across the projects in a programme
2. All project under one programme follow same rules and policies
3. Bargaining, procurement and vendor management tasks are unified
4. Risk management is centralized at the programme level
5. Information is more freely shared
6. All project managers communicate at central point i.e. to Programme manager
7. Follows common approach for stakeholder management
8. Costs and schedule are also controlled from centralized point i.e. from the programme level

2.6 Strategic Programme Management

- To carry out an effective strategic management of a project, there should be a strategic plan clearly defining the organization's objectives and programme goals. This provides the context for assessing the individual project.
- Strategic management is usually carried out by Senior Management of the organization to check if the project fits in the long-term goal of the organization. They evaluate the individual projects against the strategic plan or the overall business objectives.

Typical issues to be considered during strategic management :

- **Objectives** : How will the Proposed System contribute to the organization's objectives?
- **Organizational Structure** : What effect will the new System have on the existing organization structure?
- **MIS (Management Information System)** : What information will the System provide and at what levels in the organization.
- **Personnel** : In what way will the proposed system affect the existing employee skill base?
- **Image** : What will be the effect on customers' attitude towards the organization

2.6.1 Managing the Allocation of Resources within Programmes

- Successful strategic management ensures that all the resources in the organization perform effectively.
 - By managing the priorities, the business professionals enable the employees to balance job tasks, schedule work efficiently and ensure that work flows smoothly from one process to the next.
 - By establishing a comprehensive strategic plan to manage the allocation of the resources, you avoid costly mistakes that lead to overruns and delays.
1. Coordinate the project and operations effectively by establishing a comprehensive program management strategy.
 - o Evaluate project proposals on a monthly or quarterly basis to decide which ones gets sponsorship.
 - o Consolidate multiple similar resource efforts under one program leader who can then manage them more effectively and ensure to meet critical deadlines.
 2. Employ software tools such as Microsoft Project, dotProject.net or Base camp, to identify project tasks, allocate resources effectively, avoid over allocation and prevent employee burnout.
 3. Company executive should approve budgets, finish dates and the amount of flexibility in the deadlines to help project managers make decisions aligned with the company's strategic goals.

Delay tasks until staff has enough time to work on them or split up tasks and hire additional workers to prevent staff from working more than 40 hours per week and becoming burned out.
 4. Outsource few tasks to the companies specialized in that particular function such as payroll processing, customer service or technical support.
 5. Train employees to gain required skills so as to complete the job tasks on time to ensure timely delivery of products and services.
 - o Train less experienced workers to complete the job tasks if you experience unexpected demand or attrition.
 - o Obtain specialized training from authorized providers to ensure that your company runs a safe workplace that complies with local, state and federal regulations.
 6. Manage suppliers by analyzing work flow of resource materials from one process to the next.
 - o Gather input from experts before considering alternative solutions to backlogs.
 - o Take prompt action to rectify problems if a supplier provides poor quality materials or delivers them late.

2.7 Benefits Management

- The reason every organisation invests in projects is to capitalise on a business opportunity (e.g. to fulfil a public need or to solve an issue that the company is facing (e.g. rising admin costs).

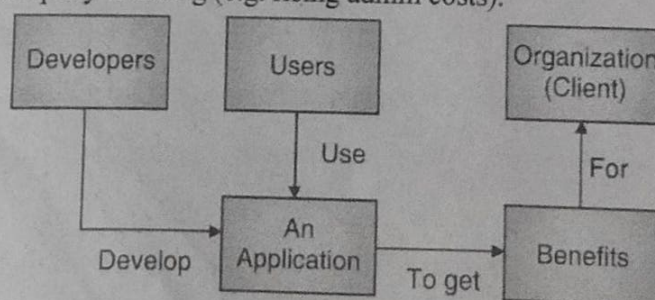


Fig. 2.7.1 : Why do organization develop an application?

- But, it is seen that huge no. of projects fail to achieve their objectives or to deliver benefits for an organisation. The causes behind this failure are when organisations fail to identify the :
 - o Issue the project is trying to address
 - o Benefits that can realistically be achieved
 - o Responsible persons who can ensure that the benefits are achieved.
- **Benefits management** is the process of identifying realistic benefits, planning, ownership, tracking and actual realisation in order to ensure that the proposed project delivers the estimated objectives and provides value for money.

Types of Benefits :

1. Direct (tangible) benefits : Benefit gained or money saved by reducing the staff and re-organizing the staff.
2. Assessable (quantifiable) indirect benefit : These are secondary benefits such as increase of accuracy through a more user-friendly screen or greater customer satisfaction.
3. Intangible benefits : These benefits are longer term in nature such as enhanced job interest leads to reduction in recruitment costs. Other examples include -corporate reputation, capacity for change.

Tangible Benefits :

1. Improved quality of service
2. Increased productivity
3. More motivated workforce
4. Better internal management and risk reduction - better decision-making. For example, in the case of an insurance company a deeper analysis of insurance claims might help to identify types of business that are most risky and thus measures to prevent these risks such as allows the company to adjust premiums to cover these risks.
5. Manage the economy i.e. cost-cutting. For example, using an automated telephone system to direct the calls without human intervention could allow to reduce the staff.
6. Revenue acceleration. For example, sooner that bills reach the customers, sooner they can pay them.
7. Strategic fit – A change should be planned in such a way that it does not just benefit a single group within an organization but benefits the organization as a whole.

Quantifying Benefits :

1. Quantified and valued – e.g. a reduction of x staff to save \$y
2. Quantified but not valued – e.g. a decrease in customer complaints by x%
3. Identified but not quantified – e.g. public approval for an organization in the locality where it is based.

2.7.1 Benefits Management Process

- The benefits of a programme or a project are forecasted to form a basis of its business case. The sponsor owns the business case and is ultimately accountable for the realization of these benefits.
- The organization identifies the benefits it needs to obtain and initiates changes to deliver those benefits. During the change, the organization monitors the performance indicators that predict the benefits delivery.
- Benefits management is an iterative process with five main steps as illustrated in the Fig. 2.7.1

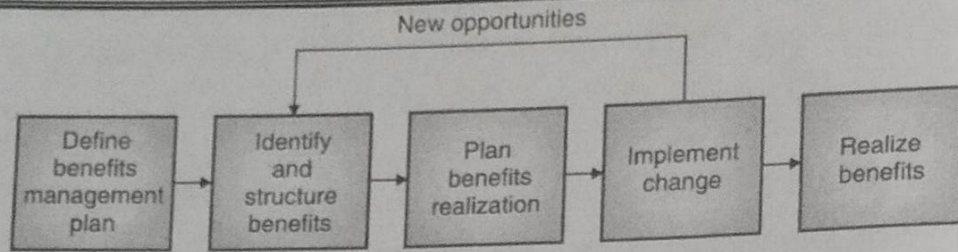


Fig. 2.7.1 : Benefits management process

- 1. Define benefits management plan :** This explains how benefits will be managed. It sets out policies for measurement, roles and responsibilities, priorities and key performance indicators (KPIs). The benefits plan illustrates the timeline and milestones for realizing benefits, including any dependencies on project outputs or interactions between benefits.
- 2. Identify and structure benefits :** Requirements are captured from sources such as the project mandates and stakeholders. Benefits depend on the delivery of outputs and the achievement of outcomes. The inter-relationships between these is understood through benefits modelling and benefits mapping. Each benefit and disbenefit should be documented in terms of priority, interdependencies, value, timescales and ownership.

Benefits mapping is a process that :

- The benefits of a programme or a project are forecasted to form a basis of its business case. The sponsor owns the business case and is ultimately accountable for the realization of these benefits.
- The organization identifies the benefits it needs to obtain and initiates changes to deliver those benefits. During the change, the organization monitors the performance indicators that predict the benefits delivery.
- Identify which benefits can be quantified and hence which benefits to measure in order to determine how successful a project has been.
- Benefits mapping should be undertaken with stakeholders who are more impacted by the project as well as those who have the capacity to make decisions e.g. senior executives. By getting sign-up at the early stage of the process there will be a common understanding of the project aims and its benefits.

Establish 'The Context'	Establish objectives of project	Identify benefits	Identify changes and enablers	Identify stakeholders	Priorities and baseline	Identify benefit measures
Identify the objective of the project	Identify Key user objectives	Identify and benefits	Identify changes and enablers required to deliver a benefit	Identify stakeholders responsible for realising the benefits	Identify gaps in the maps	Identify benefit measurements
Identify any limitations to implementing the project		Link and benefits to user objectives	Identify business Change required	Identify stakeholders responsible for realising the changes	Selected key priority benefit chains for project	Identify owner of measurement
		Identify intermediate benefits				Identify measurement plan
		Identify any disbenefits				

Fig. 2.7.2 : Benefit mapping process

Disbenefit :

- When stakeholders seek to identify benefits from a change programme, they will also identify disbenefits.



- A disbenefit is an outcome of change that is understood as negative by a stakeholder e.g. increased social exclusion.
 - Disbenefits will have a negative impact on the outcomes of the project.
 - By mapping disbenefits and their potential effects, it will be possible for the stakeholders to specify how they are going to mitigate the effects and what potentially they need to put into place to reduce the impact of the disbenefit.
 - The same benefit can be understood as a benefit to some stakeholders and a disbenefit to other stakeholders. For example, reducing staff members gives tangible benefit to organization but is a disbenefit to worker union.
3. **Plan benefits realization** : It is about capturing baseline measurement. Baseline measurements identify the current performance of an operation so that improvements can be measured.
4. **Implement change** : Benefits are achieved when something changes. This usually involves permanently changing the attitude, behaviour or physical changes.
- Identify what business changes an organisation needs to put in place in order to realise their benefits from a project – for example, a business change would be training staff to use the new computer system or introducing a new administration process. The output of this is to understand Benefit Dependency which links an organisation's objectives, changes, benefits and disbenefits together.
- **Objective** : The reason why the project is being undertaken
 - **Change** : A business change in order to realise the benefits
 - **Benefit** : Outcome of change that is seen as **positive** by a stakeholder
 - **Disbenefit** : Outcome of change that is seen as **negative** by a stakeholder
5. **Realize benefits** : The implemented changes and long-term actions should be embedded to ensure that benefits continue to be realized. A business change manager should track the realization and ensure that the change is permanent.

Benefit for a Project :

In few cases, the project ends with the delivery of the project but in few cases, the projects continue through the extended project life cycle to deliver measurable benefits. When a project is responsible only for delivering outputs, it must interface with whoever is responsible for delivering the benefits.

Benefit for a Programme :

- The benefits associated with programmes contain complex interactions between the outputs of individual projects, outcomes and benefits.
- Programme benefit can be a difficult issue particularly where investment approvals are impacted. These should be approached in a pragmatic way and resolved through effective mapping and stakeholder consultation.
- You should follow a consistent approach to benefits management across a programme to aggregate the benefits across multiple projects and assess their collective impact on business performance across the organization.

Benefit for a Portfolio :

- Strategy mapping ensures that the investment decisions and scope of each project and programme are driven by the contribution of benefits to achieve operational, organizational or business strategy.
- Portfolio must have a consistent set of guidelines for benefits management practices for all programmes and projects, including tracking, forecasting and reporting. This enables benefits to be compared and aggregated across the portfolio.



Overview of Project Planning

UNIT I

Syllabus

Introduction to Stepwise Project Planning, Step 0 : Select Project, Step 1 : Identify Project Scope and Objectives, Step 2 : Identify Project Infrastructure, Step 3 : Analyze Project Characteristics, Step 4 : Identify Project Products and Activities, Step 5 : Estimate Effort for Each Activity, Step 6 : Identify Activity Risks, Step 7 : Allocate Resources, Step 8 : Review/Publicize Plan, Steps 9 and 10 : Execute Plan/Lower Levels of Planning.

3.1 Software Project Management Plan

Project planning :

- Planning is the most time-consuming activity.
- This activity is continuously conducted right from the initial phase through the software development to system delivery.
- Plans are regularly revised so as to avail the new updates.
- Various types of plans such as for scheduling, budgeting and many such additional plans are developed so as to support the main software project plan.

Need of software project management plan :

- It is a major part of project agreement.
- It controls the software development process.
- It describes the technical and managerial concepts involved in software development.
- It is a support for requirements analysis document.

Project plan is a document that defines :

- The scope (possible functions and limitations of the system), duration (time required to develop the system), cost (estimated budget) and deliverables for the project.
- System delivery dates, delivery location. Deliverables may be :
 - o Documents – user manuals
 - o Function descriptions
 - o Non-functional requirement descriptions
 - o Components descriptions



3.2

Structure (Content) of a Software Project Management Plan

- Front matter
- Title page
- Revision sheet (update history)
- Preface : Scope and purpose
- Tables of contents, figures, tables.

1. Introduction :

- (i) **Project overview** : It involves summarized description of project and product.
- (ii) **Project deliverables** : It involves the description of items to be delivered, their delivery dates and delivery location.
- (iii) **Evolution of the SPMP (Software Project Management Plan)** : It involves descriptions of anticipated and unanticipated changes.
- (iv) **References** : It includes complete list of materials referenced in SPMP.
- (v) **Keywords, Definitions and Acronyms.**

2. Project organization :

- (i) **Process model** : Describes inter-relationships among various project elements regarding functions, activities, tasks, Milestones, Baselines, Reviews, Work breakdown structure, Project deliverables and Sign-offs.
- (ii) **Organizational structure** : Describes internal management and organizational chart.
- (iii) **Organizational interfaces** : Describes relations with other entities.
- (iv) **Responsibilities** : Describes the major functions and activities.
States responsibility of each team member as who is in-charge of which activity.

3. Managerial process :

- (i) **Management objectives and priorities** : Philosophy, goals and priorities.
- (ii) **Assumptions, dependencies, constraints** :
 - 1. Assumption may be that the security will not be addressed.
 - 2. Dependency may be that the automatic code generation facility in the CASE tool depends on JDK.
 - 3. Constraints may be that the length of the project is 3 months. Limited amount of time to build the system the project consists of beginners and so, it will take time to learn how to use the tools
- (iii) **Risk management** :
Identifying, assessing, tracking Contingencies for risks

Example :

Risk : Members in key roles drop the course.

Contingency : Roles are assigned to somebody else. Functionality of the system is renegotiated with the client

**Example :**

Risk : The project is falling behind schedule.

Contingency : Extra project meetings are scheduled.

(iv) **Monitoring and controlling :** It involves the reporting mechanism of information flows and code reviews.

(v) **Staffing Plan :** It describes the required skills needed for project management.

4. Technical process :

(i) **Methods, Tools and Techniques :** Computing system, development method, team structure, Standards, guidelines, policies.

(ii) **Software documentation :** Documentation plan, including milestones, reviews and baselines.

(iii) **Project support functions :** Plans for functions (quality assurance, configuration management).

5. Work Elements, Schedule, Budget :

(i) **Work packages (Work breakdown structure) :** Project decomposed into tasks; definitions of tasks.

(ii) **Dependencies :** Defines relations among the functions, activities and tasks in a priority.

(iii) **Resource requirements :** Lists the estimations of the resources such as development staff (personnel), development time, special hardware and support software, if any.

(a) **Budget and resource allocation :** Defines the costs related with functions, activities and tasks.

(b) **Schedule :** States the deadline, dependencies and the project milestones.

3.3 Step-wise Project Planning

Planning is the most important process in project management.

Table 3.3.1 : Step-wise Project Planning

Sr. No.	Select Project
1.	Identify project scope & objectives
1.1	Identify objectives and also identify effective practical measures to meet those objectives
1.2	Establish a project authority to ensure unity among the team members
1.3	Stakeholder analysis: identify the interests of all direct and indirect stakeholders of the project.
1.4	Modify objectives based on stakeholder analysis
1.5	Establish methods for communication with all parties
2.	Identify project infrastructure
2.1	Identify relationship between the project and strategic planning To determine the order of related projects being carried out in the organization



Sr. No.	Select Project
	To establish a framework for the system To ensure the hardware and software standards are followed
2.2	Identify installation standard and procedures related to the software project
2.3	Identify project organization
3.	Analyze project Characteristics
3.1	Distinguish the project as either objective driven or product driven
3.2	Identify high-level project risks
3.3	Consider the requirements concerning to implementation
3.4	Select a SDLC approach
3.5	Review overall estimations
4.	Identify project products and activities
4.1	Identify and describe all the products related to the project
4.2	Document the flow of the product
4.3	Recognize product instances
4.4	Produce an activity network that shows the tasks that have to be carried out in a sequence
4.5	Check compatibility of products of previous activities
5.	Estimate effort for each activity
5.1	Carry out bottom-up estimates such as staff effort, time for each activity, and other resources
5.2	Breakup the long activities into a series of smaller ones
6.	Identify activity risks
6.1	Identify the likelihood of risk occurrence and damage it incase if it occurs
6.2	Plan risk reduction activity and contingency measures
6.3	Revise overall plans and estimates considering the risks
7.	Allocate resources



Sr. No.	Select Project
7.1	Identify type of staff needed for each activity
7.2	Revise plans and estimates considering the staffing constraints and their issues
8.	Review/ Publicize plan
8.1	Review quality aspects of the project plan so as to ensure each activity is completed with a quality product each activity has 'exit requirements'
8.2	Document the plan and the commitment that is agreed upon by all the parties
9.	Execute plan
10.	Lower-level planning

Review Questions

- Q. 1 What is the need of project management plan ?
- Q. 2 What does the project plan document define ?
- Q. 3 Write short notes on content of project plan.
- Q. 4 Explain the step-wise project planning.